

Sensor-Based Nadi Pariksha Diagnostic Machine

1. Executive Summary

Nadi Pariksha is an **Ayurvedic pulse diagnosis method**. According to Ayurveda, our body is governed by three energies called **doshas**:

- **Vata** – movement, nervous system, dryness
- **Pitta** – metabolism, heat, digestion
- **Kapha** – structure, stability, lubrication

In Nadi Pariksha, a trained doctor (Vaidya) feels the pulse at the wrist and, from its **speed, strength, rhythm, and temperature**, they judge which dosha is **balanced or imbalanced**.

What your project is doing

Your project is trying to **convert this traditional manual method into a modern electronic system** by:

- Using **pressure sensors** to feel the strength and waveform of the pulse
- Using **PPG sensors** to measure **blood flow and heart rate**
- Using **temperature sensors** to measure the **surface temperature** of the wrist
- Sending all this data to a **microcontroller / Raspberry Pi**
- Writing **embedded software** to:
 - Filter noise
 - Extract important features
 - Map these features to **Ayurvedic dosha conditions**
- Displaying the result on a **mobile or desktop app** for the doctor to see

The goal is to make Nadi Pariksha:

- **Objective** – based on sensor data, not just “feel”
- **Repeatable** – same reading → same interpretation
- **Scalable** – can be used in clinics, wellness centers, and training institutes

For all of this, you are estimating a **budget of ₹6.75 lakhs**, which includes hardware, software, testing, documentation, and contingency.

2. Background: Understanding Nadi Pariksha

How Nadi Pariksha is performed traditionally

A traditional Ayurvedic practitioner:

- Uses **three fingers** (index, middle, ring) on the radial artery at the wrist of the patient’s right (or sometimes left) hand.
- Each finger is placed at a **slightly different position** along the artery.
- Each finger corresponds to one **dosha**:
 - **Vata – Index finger**
 - Feels like a *snake* (thin, quick, irregular)

- Indicates air and space elements, movement, nervous activity
- **Pitta – Middle finger**
 - Feels like a *frog* (jumping, strong, warm)
 - Indicates fire and water elements, metabolism and heat
- **Kapha – Ring finger**
 - Feels like a *swan* (slow, deep, smooth, steady)
 - Indicates earth and water elements, stability and heaviness

Parameters checked by practitioner

The Vaidya is not just counting heart rate. They are “feeling” many things:

- **Vega (Rate)** – How fast is the pulse? Faster pulses may indicate Pitta or Vata aggravation; slow and heavy pulses may indicate Kapha.
- **Tala (Rhythm)** – Is it regular like a metronome? Or irregular, skipping beats, varying rhythm?
 - Regular rhythm → more balanced
 - Irregular → Vata disturbance or pathology
- **Bala (Strength/Force)** – Is the pulse very strong and hard, or weak and feeble?
 - Strong, bounding → Pitta or strong circulation
 - Weak → Vata or low vitality
- **Tapamana (Temperature)** – Is the wrist area hot, warm, cool, or cold?
 - More warmth → Pitta dominance
 - Coldness → Vata or poor circulation

Why digitization is needed

Traditionally:

- Everything is **subjective** – depends on the practitioner’s touch
- Two practitioners may **interpret differently**
- There is no stored data or graph
- Students cannot “see” what the teacher is feeling

By digitizing:

- We can **record** pulse waveforms
- We can **store** data for comparison over time
- We can **train students** using real signals
- We can gradually build **datasets** linking sensor signals to Ayurvedic diagnosis

So the background section explains **what Nadi Pariksha is** and **why it makes sense to map it to sensors**.

3. Problem Statement

Even though Nadi Pariksha is powerful and non-invasive, it is **not widely used** in mainstream modern healthcare. The reasons:

1. **Heavy dependence on practitioner experience**
 - Only highly experienced Vaidyas can perform it correctly.
 - Training takes many years, and still results may vary person to person.
2. **Lack of standardization**
 - There is no standard device to measure pulse features.
 - No standard scale (e.g., Bala = 0–10) or objective parameters.

3. **No supporting digital tools**
 - Modern doctors and students don't have devices that show pulse waveforms linked to doshas.
 - Teaching remains mostly verbal and demonstrative.
4. **Limited integration with telemedicine and public health**
 - In telemedicine, you need **digital input** from patient side.
 - Without sensors, Nadi Pariksha cannot be done remotely.

What your project solves

Your device aims to:

- **Digitize** the pulse signals
- Provide **consistent measurements** for all patients
- Assist practitioners in making decisions based on **both**:
 - Traditional Ayurvedic knowledge
 - Objective biomedical signal features

So the problem statement is clearly:

“Nadi Pariksha is valuable but underused because it is subjective, non-standard, and lacks digital support. Our project will fix this by creating a sensor-based, standardized diagnostic device.”

4. Project Objectives – Detailed

You listed concise objectives; here's what each really means:

1. **Designing a wearable device with multiple biosensors**
 - A wrist-mounted housing that holds:
 - 3 pressure sensors (aligned to three dosha positions)
 - 3 PPG sensors (aligned to corresponding points)
 - 3 temperature sensors
 - It should be comfortable, adjustable, and stable during measurement.
2. **Capturing and digitizing real-time radial pulse data**
 - Convert analog signals from sensors to digital using an **ADC**
 - Use Raspberry Pi or microcontroller as the processing hub
 - Sample signals at appropriate frequency (e.g., 100–500 Hz)
3. **Mapping pulse characteristics to Ayurvedic parameters**
 - Example:
 - High rate + strong amplitude + warmth → Pitta aggravation
 - Irregular rhythm + weak amplitude → Vata imbalance
 - Slow, deep, strong but cold → Kapha dominance
 - These mappings are built with Ayurvedic expert help.
4. **Interpreting doshic patterns using software algorithms**
 - Use a combination of:
 - Thresholds and rules (if HR > X and temp > Y → Pitta)
 - Machine learning later (if enough data collected)
5. **Developing an intuitive interface**
 - For doctors:
 - Live waveform
 - Dosha dominance meter (e.g., bar graph Vata/Pitta/Kapha %)
 - Basic suggestions (e.g., “Vata aggravated; consider lifestyle correction”)
6. **Validating results through trials & expert feedback**
 - Take readings of real patients

- Ask Ayurvedic doctor to record their diagnosis manually
 - Compare and refine algorithms until they match reasonably well
7. **Preparing documentation for scaling and production**
- Proper design documents, BOM, circuit diagrams, testing reports
 - So that later it can be manufactured or submitted for grants/approvals.
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5. Components of the System

A. Sensors and Hardware – Explanation

Let's explain **why each component is chosen**:

- 1. Piezoelectric Pressure Sensors (3 units)**
 - These convert **mechanical pressure** (pulse beat) into **electrical voltage**.
 - Placed under index, middle, and ring finger positions.
 - They generate a waveform corresponding to **pulse pressure over time**.
 - From this, we can calculate:
 - Pulse amplitude
 - Shape, upstroke, downstroke
 - Strength (Bala)
- 2. PPG (Photoplethysmography) Sensors (3 units)**
 - These use a **light source (LED) and detector**.
 - As blood volume changes with each heartbeat, light absorption changes.
 - PPG helps in:
 - Heart rate
 - Heart rate variability (HRV)
 - Blood flow pattern
 - It is commonly used in smartwatches and oximeters.
- 3. IR Thermopile Temperature Sensors (3 units)**
 - These detect **infrared radiation** from the skin surface.
 - Gives **non-contact or slight-contact skin temperature** above radial artery.
 - Helps to quantify **Tapamana** (heat or coldness).
- 4. Raspberry Pi / ADC Board (1 unit)**
 - Acts as main controller and mini-computer.
 - Reads signals via ADC (if external)
 - Runs Python/C programs to:
 - Filter signals
 - Extract features
 - Display results
- 5. Signal Amplifiers & Filters (Set)**
 - Raw sensor signals may be:
 - Weak (microvolts/millivolts)
 - Noisy (from body movement, environment)
 - Amplifiers boost signal.
 - Filters (Butterworth, etc.) remove unwanted frequencies (e.g., >50 Hz noise).
- 6. Rechargeable Battery & Power Pack**
 - Makes it **portable** – can be used in clinics without depending on mains power.
 - Supports hours of use.
- 7. 3D Printed Wrist Enclosure (2 units)**
 - Holds all sensors in **fixed, repeatable positions**.
 - Ensures the same alignment for every patient.
 - Two units may be for two sizes or as backup.
- 8. Custom PCB Fabrication (2 boards)**

- Instead of breadboards and loose wires, a **PCB** is more stable and professional.
- Integrates:
 - Sensor connectors
 - Amplifier ICs
 - Filter circuits
 - ADC interface paths

9. Connectors, Spares, Wiring

- For internal connections, sensor replacement, debugging.

10. Costing and Contingency

- Subtotal: ₹97,500 (raw hardware)
- Contingency 25% for:
 - Damaged sensors
 - Design revisions
 - New components needed
- Final hardware cost: ₹1,21,875

B. Software and Processing – Detailed

Your embedded software will:

1. Acquire signals

- Read data from pressure, PPG, temperature sensors
- Use specific sampling frequency per sensor
- Time-align the readings from all three dosha positions

2. Filter and clean raw signals

- Use **Butterworth filters** because they have:
 - Flat frequency response in passband
 - Predictable behavior
- Low-pass filter: removes high-frequency noise
- Band-pass filter: keeps only frequency range of pulse (around 0.5–20 Hz)

3. Extract features

From pressure & PPG:

- **Heart Rate (HR)** – count beats per minute
- **Heart Rate Variability (HRV)** – variation in time between beats
- **Waveform amplitude** – strength of pulse
- **Slope of rising and falling edges** – indicates vascular condition
- **Rhythm consistency** – regular vs irregular (for Vega & Taal)

From temperature:

- **Average surface temperature**
- Trend (rising, stable, falling)

4. Interpret Dosha Imbalances

Examples:

- Fast HR + strong amplitude + high temperature → **Pitta high**
- Irregular HR + low amplitude + maybe cold periphery → **Vata high**
- Slow HR + heavy/strong/deep wave + cool or clammy → **Kapha dominance**

These rules will be refined with **expert Ayurvedic input**.

5. Display visualizations

- Show:
 - Live pulse graphs

- HR, HRV, temperature values
 - Dosha bar graph or pie chart
 - Text interpretation like:
 “Predominant dosha: Vata. Signs of Vata aggravation observed in pulse characteristics.”
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C. Testing and Validation

To make your device **scientifically credible**, you must prove that its output **matches expert diagnosis**.

Steps:

1. **Select 20–30 subjects**
 - Different ages, genders, health conditions.
 2. **For each subject**
 - Place the device on the wrist and record signals for a fixed duration (e.g., 2–3 minutes).
 - At the same time, an experienced Ayurveda doctor performs manual Nadi Pariksha and notes:
 - Dominant dosha
 - Any imbalance (e.g., “Vata-Pitta aggravation”)
 3. **Data comparison**
 - Compare the device’s dosha reading with the doctor’s reading.
 - Calculate:
 - Percentage of agreement (e.g., 80% match)
 - Where it differs and why
 4. **Refinement**
 - Adjust thresholds in algorithm based on feedback.
 - Test again until reasonable accuracy is achieved.
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D. Documentation, IP & Outreach – Detailed

You will prepare:

- **User Manuals**
 - How to place the device correctly
 - How to interpret software results
 - Safety precautions
- **Training Guides**
 - For students learning Nadi Pariksha with both hand and device
 - Step-by-step tutorials with example recordings
- **Clinical Whitepaper**
 - A documented study of your trials:
 - Methodology
 - Data analysis
 - Results
 - Discussion
- **Provisional Patent**
 - For:
 - Integration of multiple sensors for Nadi Pariksha
 - Custom algorithms mapping signal to doshic interpretation
- **Outreach**
 - Live demos at Ayush centers, wellness clinics, Ayurveda colleges

- Collect feedback from practitioners
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E. Contingency

This is a buffer budget for:

- Sensor failures
 - Extra PCBs
 - Extra 3D prints
 - Travel for clinical testing/demo
 - Any redesign if first prototype fails in some aspect
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6. Budget Summary – Detailed

You have divided your total cost into 5 parts:

1. **Hardware – ₹1,21,875**
 - All sensors, electronics, enclosure, and contingency.
2. **Software – ₹1,50,000**
 - Developer time for:
 - Firmware programming
 - Signal processing algorithms
 - UI/UX design for interface
 - Possibly small licenses / software tools.
3. **Testing – ₹1,20,000**
 - Honorarium for Ayurvedic experts
 - Clinical site costs
 - Subject recruitment, printing, logistics
4. **Documentation & IP – ₹85,000**
 - Printing manuals, design reports
 - Patent filing costs (provisional)
 - Whitepaper preparation, graphics, diagrams
5. **Contingency – ₹50,000**
 - Final buffer for any unexpected expense

Total = ₹6,75,000

This shows you have **planned a realistic budget**, not just for hardware but also for **software, testing, and legal/IP work**, which many student projects ignore.

7. Timeline

You've divided the work into **5 phases over 9 months**:

1. **Phase 1 (1.5 months)**
 - Finalize component list
 - Purchase sensors and electronics
 - Design and fabricate PCB
 - Design 3D wrist enclosure
 - Basic bench testing of sensors
 2. **Phase 2 (2 months)**
 - Implement data acquisition code
 - Implement filters (Butterworth etc.)
 - Design and code the UI (desktop/mobile)
 - First version of dosha interpretation algorithm
 3. **Phase 3 (2 months)**
 - Start clinical testing with small group of subjects
 - Collect parallel readings (device + doctor)
 - Analyze mismatch cases
 - Refine algorithms and recalibrate sensors
 4. **Phase 4 (2 months)**
 - Deploy device in one pilot Ayurveda center or wellness clinic
 - Collect real-world feedback from practitioners and patients
 - Improve ergonomics, UI, stability
 5. **Phase 5 (1.5 months)**
 - Prepare final documentation
 - Write clinical whitepaper/report
 - File provisional patent
 - Prepare presentations for funding agencies / colleges / competitions
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8. National Relevance

Your project aligns with multiple **Government of India initiatives**:

1. **Ministry of AYUSH**
 - Ministry promotes **Ayurveda, Yoga, Unani, Siddha, Homeopathy**.
 - Your device helps **standardize an Ayurvedic diagnostic method**, which fits AYUSH goals perfectly.
2. **Ayushman Bharat – Health & Wellness Centres**
 - These centers focus on **preventive healthcare**.
 - Your device can be used for:
 - Early detection of imbalances
 - Lifestyle counseling
 - Non-invasive, low-cost screening
3. **National Digital Health Mission (NDHM)**
 - NDHM promotes **digitized health records (ABHA ID)**.
 - Your device can store patient pulse data and link it with their **ABHA health ID**, creating a digital history of their doshic patterns.
4. **Startup India / Atal Innovation Mission**
 - The project has **startup potential**:

- Indigenous hardware
- Ayurveda-specific digital health device
- It can be scaled:
 - Sold to clinics, Ayurvedic colleges, wellness chains
 - Exported as a unique Indian innovation

So you are not just doing a college mini-project; you are creating a **product concept that fits into national health and innovation frameworks.**

9. Conclusion

To summarize in an expanded way:

- **Traditional Nadi Pariksha** is a powerful but subjective diagnostic tool in Ayurveda.
- Modern healthcare systems demand:
 - **Objectivity,**
 - **Standardization,**
 - **Digital records,** and
 - **Scalability.**

Your **Sensor-Based Nadi Pariksha Diagnostic Machine:**

- Combines **Ayurvedic wisdom** with **modern sensors and embedded systems.**
- Uses:
 - Pressure sensors (for Bala and waveform),
 - PPG sensors (for HR and HRV),
 - Temperature sensors (for Tapamana),
 - Raspberry Pi / microcontroller (for processing),
 - Algorithms (for dosha interpretation).

With a carefully planned **budget of ₹6.75 lakhs** and **9-month timeline**, the project is:

- **Technically feasible**
- **Culturally relevant**
- **Aligned with national healthcare policies**

You finally **request financial support** to:

- Build the prototype
- Validate it with expert Ayurvedic practitioners
- Document the findings
- Move towards a scalable, production-ready health device